Case Study III: Movie Recommender

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Problem Introduction

Building a movie recommender with **LensKit**

LensKit makes it easier to train and test recommendation algorithms

Worked with the **MovieLens 32M dataset**

Goal: design a simple interactive **movie recommender**

Streamlit used to create dashboard

Compared recommender system to existing ones

Analysis of the Problem

- Information on Title, Genre, Rating, Tags
- TMDB Data available -> Movie Posters
- Assumed that Movie Popularity would be important to users

Approach



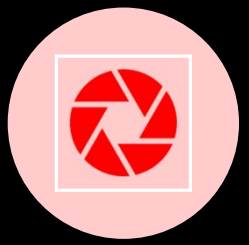
Content-Based



Collaborative Filtering

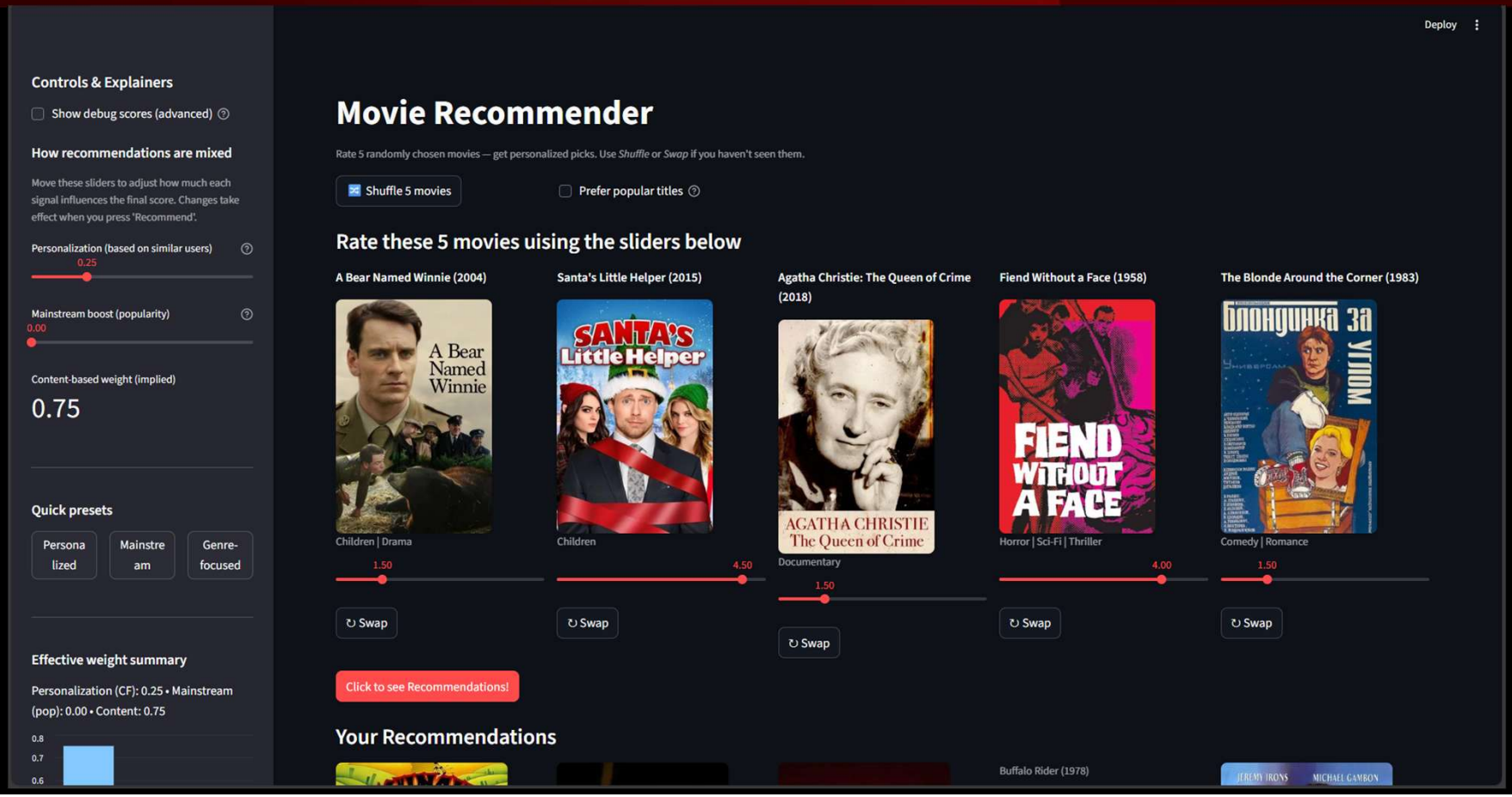


Popularity Baseline



Hybrids + Diversity

Findings


Deploy
Controls & Explainers
Show debug scores (advanced)
How recommendations are mixed
Move these sliders to adjust how much each signal influences the final score. Changes take effect when you press 'Recommend'.
Personalization (based on similar users)
0.25
Mainstream boost (popularity)
0.00
Content-based weight (implied)
0.75
Quick presets
Persona lized
Mainstre am
Genre-focused
Effective weight summary
Personalization (CF): 0.25 • Mainstream (pop): 0.00 • Content: 0.75
0.8
0.7
0.6
Movie Recommender
Rate 5 randomly chosen movies — get personalized picks. Use Shuffle or Swap if you haven't seen them.
Shuffle 5 movies
Prefer popular titles
Rate these 5 movies uising the sliders below
A Bear Named Winnie (2004)
A Bear Named Winnie
Children | Drama
1.50
Swap
Santa's Little Helper (2015)
SANTA'S Little Helper
Children
4.50
Swap
Agatha Christie: The Queen of Crime (2018)
AGATHA CHRISTIE The Queen of Crime
Documentary
1.50
Swap
Fiend Without a Face (1958)
FIEND WITHOUT A FACE
Horror | Sci-Fi | Thriller
4.00
Swap
The Blonde Around the Corner (1983)
Comedy | Romance
1.50
Swap
Click to see Recommendations!
Your Recommendations
Buffalo Rider (1978)

Findings

Deploy

Your Recommendations



Shawshank Redemption, The (1994)

score: 0.693



Good Will Hunting (1997)

score: 0.678



Sixth Sense, The (1999)

score: 0.677



Matrix, The (1999)

score: 0.666



Back to the Future (1985)

score: 0.663



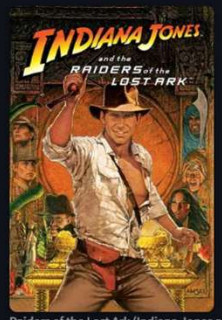
Forrest Gump (1994)

score: 0.662



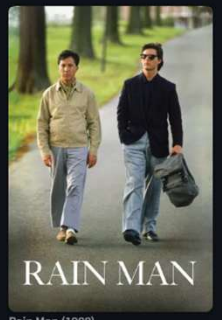
As Good as It Gets (1997)

score: 0.657



Raiders of the Lost Ark (Indiana Jones and the Raiders of the Lost Ark) (1981)

score: 0.657



Rain Man (1988)

score: 0.657



Silence of the Lambs, The (1991)

score: 0.655

Download CSV

Verification & Validation

- Functional checks
- Testing of recommender

- Problem goal
- Controls match user intent

Conclusions

- A working movie recommender + interactive **Streamlit** dashboard
- Personalization controls
- 10 tailored recommendations
- Comparison to Netflix and Spotify
- User friendly and intuitive

Recommendations



Deeper Personalization



User accounts



‘Already watched’ option